Evaluating Bay Area Equity Priority Communities' Commuting Mobility, Utilizing Daytime Population Data from 2023

GitHub Link

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Track B

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1.0 Problem and Motivation

Research question

The Metropolitan Transportation Commission defines Equity Priority Communities (EPCs) as "census tracts that have a significant concentration of underserved populations, such as households with low incomes and people of color." Within these census tracts, the type of work and typical daily commute may differ compared with the general working population. Our team aims to explore:

How does Bay Area average daily commuting mobility vary between EPCs and non-EPCs, using daytime population data from 2023?

To understand these patterns from a wider perspective, we will conduct our analysis and visualization using the following guiding questions:

1. How does the net change in daytime commuters in EPCs compare to non-EPC tracts?
2. How does the percentage of commuters in EPCs compare to surrounding census tracts?
3. Does the average resident in an EPC travel *further distances* and for *longer periods* of time to work than the average resident of the surrounding census tracts and neighbourhoods?
    a. By private auto
    b. By transit

Geographic Bay Area: defined by the Association of Bay Area Governments (ABAG) as the following 9 counties: Alameda, Contra Costa, Marin, Napa, San Francisco, San Mateo, Santa Clara, Solano, and Sonoma. Our data will be filtered using this geographic definition.

Bay Area Equity Priority Communities (EPCs): defined geographically by the Metropolitan Transportation Commission. Our analysis encompasses working residents of EPC and non-EPC areas that are captured by LODES data.

Daytime population: defined by the US Census Bureau as "the number of people who are present in an area during normal business hours, including workers". This contrasts with the 'resident' population present during the evening and nighttime hours.

Stakeholders and impact

The primary stakeholders are the municipal governments of the Bay Area, MTC, and EPC residents. The intentional impact of this project is to provide a tool that the public and policymakers can interact with to understand how EPCs and their neighboring tracts have different commuting travel patterns. Understanding these differences can help improve access to employment and/or transportation opportunities for EPC residents. Other groups who may benefit from this analysis include transportation providers such as transit agencies and micromobility providers. Transportation providers may choose to pilot new lines or services based on the transit accessibility in EPCs between home and work tracts.

Unit of analysis

The project team is analyzing the research question at the tract level, as EPCs are defined at the tract level. ACS 5-year data is also provided at the tract level. LODES data comes in at the block level but can be easily aggregated to the tract level, which helps to reduce noise and ensures the synthetic LODES data closely matches observed distributions [x].

Measures and metrics

To answer our first research question, we plot the daytime, nighttime, and net change for each tract, using population counts. To answer our second research question, we use a measure of the percentage of commuters in EPCs versus their direct neighbours. For our third research question, we use driving and transit distance time as a measure to determine if EPCs commute for longer periods of time compared to their neighbours.

Each of these measures will be plotted in interactive maps, histograms, or other visualization tools. Instead of assigning a metric to determine if the non-EPC tracts are better off, users can use these tools to determine the difference between each measure in EPCs and non-EPCs.

2.0 Data and Cleaning

Name	Coverage	Granularity	Type	Format	Source
LODES8	Bay Area	Census Block, annual (2023), aggregated	Numerical and Categorical	Tabular	US Census Bureau
ACS 5-Year Estimates	Bay Area	Census Tract, 5 years (2019-23), aggregated	Numerical and Categorical	Tabular	US Census Bureau
EPC Dataset and Shapefile	Bay Area	Census Tract	Numerical and Categorial	Tabular and vector	MTC/ABAG

The full cleaned datasets, and data clean code, can be found in our project github.

2.1 LODES8 Data - Home and Work Tract Pairs

For the LODES8 California 2023 OD dataset, there weren't many issues with the data. The dataset was checked for duplicates, but none were found. The GEOID columns were cast as strings so that no issues would come up. The column naming convention was already very readable, so nothing was done to change that. A column was created that took the full GEOID and sliced it to include only the state and county parts. This was then used to filter for the 9 Bay Area counties as defined earlier in this report. Another column was created that took the full GEOID and sliced it to include the state, county, and tract. This was done to help with filtering in future functions. The data was then saved to a parquet file.

LODES data is at the census block level, while ACS data is at the census tract level. Aggregating LODES data to the tract level may lose some granularity; however, findings regarding EPCs will still be applicable as EPCs are defined at the tract level.

To determine the daytime and nighttime population, the LODES data was grouped and summed by home and work tracts. A “within_tract” column contains the count of individuals who remain within their home tract for work.

2.2 ACS Data - Time of Departure to Work

The ACS 5-Year Estimates tables already come partially cleaned and ordered, so there weren’t many issues or cleaning steps necessary. Data from ACS for our counties of interest (Geographic Bay Area, as defined above) at the tract level was loaded via api call, and a copy of this raw data was saved as a csv file.

To begin the cleaning process, the ACS tables were filtered to include only the data for Total Population (ACS table B01003) and Time of Departure to Work (ACS table B08302). Sanity checks were performed to make sure the values from each of the departure time groups sum to the total number of commuters in a given tract. The cleaned data was then saved as a csv file.

The cleaning process was as follows;

1. Column names were changed from the ACS variable numbers (eg. B08302_008E) to a more understandable naming convention, describing the departure time brackets in military time, and labelling margin of error columns ‘moe’ as shorthand.
2. Datatypes were converted from floats to integers for the departure time count estimates, and from objects to strings for the name, GEOID, state, county, and tract columns.
3. Finally, the GEOIDs were shortened to state-county-tract to match the LODES8 GEOID columns for future filtering and merging.

2.3 EPC Data and Geometries

EPC shapefiles from MTC [x] were cleaned by removing columns that are unnecessary for our analysis and renaming columns to a more readable format. No duplicates or NaN values were present in the dataset.

Census tract geometries from the Census Bureau [x] were filtered for the nine Bay Area counties and plotted to ensure accuracy. The dataset contained no NaN values, and the CRS matched the EPC data.

3.0 Missing data analysis

Missing data includes some types of employment data: LODES data does not cover all types of employment (eg. gig workers, cash jobs) which could result in underrepresentation in EPCs. This falls under DD Type 1 - Data We Know Are Missing. Although we cannot do anything to mitigate this missing data, we will acknowledge it in our report and github page to make viewers aware of the bias present in our data. A second type of missing data is that our data (LODES and ACS time of departure to work) only includes workers over the age of 16 who commute to work. This is sufficient for our analysis, but does not present a complete picture of EPCs and equity considerations. This falls under DD Type 3 - Choosing Just Some Cases. The

visualization of the percentage of commuters by tracts highlights the difference between the total population and the working population captured by LODES.

4.0 Initial results & visualizations

An initial comparison of daytime vs nighttime populations by county was conducted to determine which counties have major employment hubs, and to highlight patterns at the county level. The results show that San Francisco and Santa Clara counties have the largest discrepancies between nighttime and daytime population - as expected, these are two counties with large employment hubs. The daytime and nighttime population by county is shown in Figure 1.

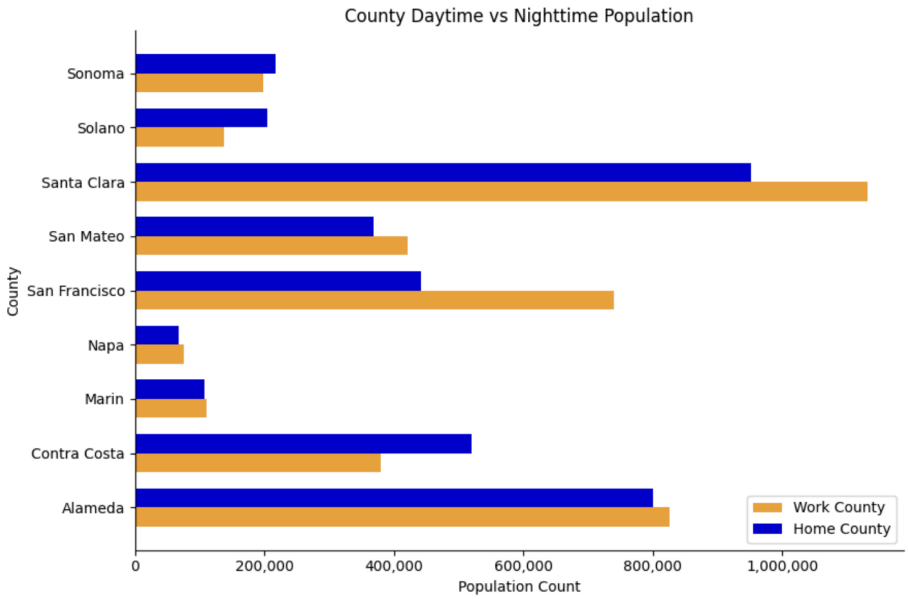
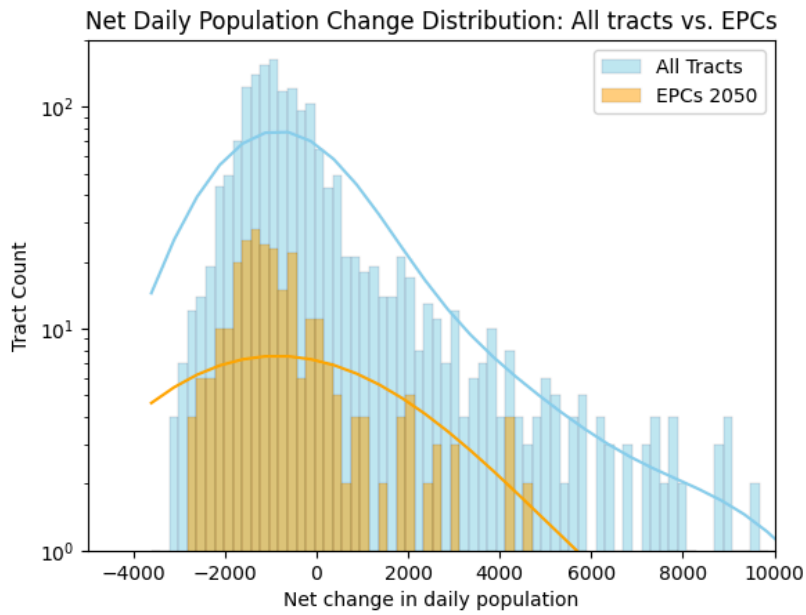


Figure 1: Daytime vs Nighttime Population in Nine Bay Area Counties

We have created four interactive map visualizations of the (a) percentage of total tract population who are commuters, (b) daytime commuter population, (c) nighttime commuter population, and (d) net change in commuter population in the Bay Area. These are all available in html in our project github. Maps (b) and (c) show the population magnitude via a color hue scale, with light color depicting a small population and dark color depicting a large population. Map (d) similarly utilizes a color scale, but this time with diverging colors to represent positive and negative net change from night to day. All three maps have EPC census tracts overlaid with an outline, chosen to be in a color that contrasts against the underlying hue scale, taking into account colorblind accessibility.

To further compare the difference in commuters between EPCs and and the wider Bay Area, we plotted the distributions of net daytime commuter change for EPC tracts and all tracts, as shown in the figure below. A log scale was used for the y-axis to see the trend visually better.



5.0 Plan to Completion

The overarching research question remains the same; however, some changes were made to the sub-questions. The last sub-question concerning commute affordability and cost was removed to reduce the scope of the project. Question 3 regarding public transit performance was redefined to be more measurable. The data sources remain the same.

Originally, the project team planned to create a 24-hour animation of population occupancy in each tract in the Bay Area over time, but after evaluating the research question further, it was found that this was not as relevant as other visualizations the team is planning to create, so it was not pursued.

Remaining Work

- Using OSMnx, find the weighted average travel distance and time by private auto for EPC and non-EPC tracts
- Using R5, find the average travel distance and time by transit for all tracts
- Develop a map of average travel distances and times for car and transit
- Examine the difference in commuting distances between EPC and non-EPC communities

Timeline to Final Submission

April 3	April 14	April 21	April 28	May 4	May 10
Submit project midterm	Visualizations to answer Question 2 and Question 3A completed	Visualizations to answer Question 3B completed	First draft of visualization GitHub Pages	Final report first draft completed	Submit final draft